

Types Without Taxa

A Covariance-Matched-Null Multiverse Test of Categorical versus Continuous Personality Structure

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2026-06-25

Abstract

A clustering method applied to personality data will return some number of “types” whether or not real types exist, which makes the familiar announcement that a dataset contains, say, four personality types hard to read: the number may describe people, or only the method. This paper offers a way to separate the two and applies it to the personality case. I compare each of three large public datasets (roughly one million respondents, across the IPIP-NEO-120, an IPIP Big-Five inventory, and the IPIP-HEXACO) against a matched null: a synthetic population built to reproduce the data’s trait distributions and full correlation structure exactly while containing no types by construction. The comparison runs across all fourteen covariance parameterizations a standard Gaussian mixture model can take, and a taxometric analysis provides an independent second reading. The full procedure was preregistered and run on held-out data. The number of types selected ranges from one to ten as a function of the covariance assumption alone; by the preregistered statistic it never exceeds what the typeless null produces; and no trait reaches a categorical threshold. The real data depart from the null only in higher-order structure, without acquiring types or turning categorical. Dimensional structure in personality is already well established; the contribution here is the test, a procedure to which any future claim of N types can be held.

Introduction

Few ideas about personality are as easy to like as the idea that people come in kinds. We call each other introverts and extraverts as though naming two species; we sort ourselves into four-letter codes; and every few years a study announces that the data, left to themselves, fall into a handful of personality types. One widely cited paper found four (Gerlach et al. 2018). The appeal is no mystery. A type is something you can hold in your head; a position in a five- or six-dimensional space is not.

But a clustering method will find types whether or not any are there. Give a mixture model a single smooth cloud of data and it still reports two or three or ten groups, because each added group trims the fit a little and the selection rule takes the trade. A clean partition, on its own, establishes little. The question is whether its groups mark anything beyond the shape the data already had.

On personality itself, the argument is largely over. Taxometric¹ and structural studies have looked for categories in trait data for years and kept finding gradients instead (Haslam et al. 2012, 2020). I do not aim to reopen that. What the settled view still lacks is a way to check a specific claim, this dataset and these respondents and this many types, against a null that means something, and a sense of how much the claim depends on choices made before any result is in view.

Past attempts to impose that discipline fall short in ways worth naming. The best-known typology study tested its clusters against data with all the associations shuffled out (Gerlach et al. 2018), but

¹ A *taxon* (plural *taxa*; from Greek *táxis*, “arrangement”) is the term of art in taxometrics (Meehl 1995; Ruscio et al. 2006) for a genuine latent category, a discrete kind of person as opposed to a position on a continuum. The title’s contrast is between *types* that a clustering method reports as output and *taxa* that would have to exist in the population for those types to refer to anything real.

data with no correlations is a low bar, and almost anything real clears it. A later comment said as much, without building the alternative it asked for (Katahira et al. 2020). Taxometrics asks the categorical-versus-continuous question head-on (Meehl 1995; Ruscio et al. 2006), but it works from a fixed set of indicators and has nothing to say about how many clusters there are, or how that number moves when the model changes.

My approach has three parts, and none is new on its own. The null is a Gaussian copula. It keeps every trait's distribution and the whole covariance matrix and scrambles only what is left, the higher-order dependence, so a cluster counts as real only if it survives a world that already shares the data's margins and correlations. This is the constructed null the earlier comment called for. Rather than report a single number, I run the same test across all fourteen covariance parameterizations of the mixture model and report the full spread of answers in place of any single preferred one. And I bring in taxometrics as an independent check, so the verdict does not rest on one method's assumptions. All of it is aimed at the personality-types question, across three datasets.

The picture that comes back is consistent. Change the covariance assumption and the number of types walks from one to ten. Hold the rest fixed and, by the measure I committed to in advance, that number is no different from what the typeless null gives. No trait crosses the taxometric line. None of that is new about personality. What it shows is that the test works, and that the next typology has something to answer to.

Background

A cluster solution settles nothing on its own

Sorting people into kinds recurs in quantitative psychology under one label after another: clinical subtypes, learner profiles, market segments, data-driven personality types. The trouble is the same each time. Clustering and mixture methods exist to partition; give them continuous data and they partition it anyway, and a selection criterion will almost always prefer more than one piece. So “the data support k clusters” quietly mixes a fact about people with a fact about the procedure, one that, left to itself, never says what a world with no types would have produced under the same treatment.

What is already settled

On the substance, the verdict is in: two meta-analyses of the taxometric literature find trait structure dimensional, not categorical (Haslam et al. 2012, 2020). What that settled verdict still lacks is a tool, one that turns a particular cluster-count claim into something that can fail, and shows how much it leans on choices no one inspects.

The null is where the argument lives

An apparent grouping is “real” only against a null that says what real means here. Shuffle every association out of the data (Gerlach et al. 2018) and you have set the bar on the floor; the correlations alone will clear it. Katahira et al. (2020) saw this but stopped at the diagnosis. The cure is borrowed from elsewhere, from the surrogate-data and matched-null traditions² of nonlinear time series, net-

²The surrogate-data method comes from nonlinear time-series analysis (Theiler et al. 1992; Schreiber and Schmitz 2000): fix the innocent features of a signal, here the margins and the linear covariance, randomize

work neuroscience, and ecology, where you fix everything innocent about a dataset and randomize only the thing in question. For a claim about clusters the copula null is the natural choice: keep the margins, keep the covariance, and let only higher-order dependence vary, so any extra clustering has to come from structure the null cannot hold.

A second opinion

Taxometrics reaches the same question by a different road, asking whether the covariation among a construct's indicators looks more like one dimension or two latent groups (Meehl 1995; Ruscio et al. 2006). The Comparison Curve Fit Index scores the observed curves against curves simulated from frankly dimensional and frankly categorical data. Because its assumptions are not a mixture model's, it makes a useful outside check, though it works on a fixed indicator set and says nothing about how many clusters there are.

Reporting the spread, not a point

That a result can hang on a defensible choice is the premise of multiverse and specification-curve analysis (Steege et al. 2016; Simonsohn et al. 2020), with a longer root in the sensitivity arguments of econometrics (Leamer 1983). The honest move is to run the reasonable specifications and show the distribution instead of defending one. For the number of types, the choice that matters is the mixture model's covariance assumption, which I vary in full; the matched null then says where, on that spread, a typeless world would have landed.

only the feature under test, and ask whether the real statistic exceeds its surrogate distribution. The Gaussian-copula null carries that logic to a mixture-model count.

Method

Design and transparency

Everything that could push the result toward “types” or “no types” was fixed before I saw the test data. Before opening the held-out data I registered the decision rule and the specification grid (OSF, seed 20260620), and what follows reports every dataset, every covariance model, and the whole rule, with nothing added or dropped once the hold-out was in view.

Data

I reanalyze three large public datasets, each already scored to traits and, where the instrument allows, facets: the IPIP-NEO-120 (Johnson 2014) (410,376 respondents; five domains, thirty facets); an IPIP fifty-item Big-Five set (Goldberg 1999) (603,322; five domains, no facets defined); and the IPIP-HEXACO (22,734; six domains, twenty-four facets). Three instruments rather than one, so that a verdict has to repeat across measurement systems instead of surviving in a single one.

I split each dataset in half (seed 20260620) into an exploratory and a confirmatory part, and sealed the confirmatory part before fitting anything reported here. For the NEO-120 and the fifty-item set the held-out respondents are almost all unseen, about 95 to 97 percent; HEXACO is small enough that the halves overlap, with about 88 percent of its hold-out also in the exploratory split, so I treat it as a robustness check rather than a clean confirmation. Every model was fit to a random subsample of the held-out part, 8,000 cases at the domain level and 5,000 at the facet level. One housekeeping point matters later: mixtures tend to select more components as the sample grows, so I always compare the real data and its null at the same sample size.

The covariance multiverse

A mixture model asks how many components best describe a cloud of points, and the answer turns on a setting most studies fix without comment: the shape the components are allowed to take. I model each dataset as a Gaussian mixture and let “types” mean the components a BIC selection keeps. `mclust` (Scrucca et al. 2016) offers fourteen ways to constrain a component’s covariance, in volume, shape, and orientation, running from the spherical (EII, VII) through the diagonal (EEI, VEI, EVI, VVI) to the fully ellipsoidal (EEE through VVV).³ For each, I fit one to ten components and record the number BIC prefers. The fourteen answers, taken together, are the specification curve: the number of types found as a function of an assumption that carries no meaning about people.

The Gaussian-copula matched null

To tell a real clustering from one that any dataset of this shape would show, I compare it against a Gaussian-copula null:⁴ a twin of the data with the same margins and the same correlations, but no types built in. The null holds those two things fixed and frees the rest, replacing all higher-order dependence with plain Gaussian dependence, so by construction it contains no latent types.

³ The three-letter codes name how each component’s covariance is held in volume, shape, and orientation: E for equal across components, V for varying, I for the identity (a spherical or axis-aligned shape). EII is equal-volume spherical; VVV lets volume, shape, and orientation all vary freely (Scrucca et al. 2016).

⁴ The term *copula* (Latin *copula*, “tie, link,” the root of English *couple*) is borrowed in statistics from Sklar (1959), where it denotes the function that ties a set of one-variable marginals into a joint distribution, separately from the marginals themselves. Any multivariate distribution decomposes uniquely into its marginals and a copula; the *Gaussian* copula is the one whose dependence is fixed entirely by a correlation matrix, with no further joint structure.

A useful way to picture this null is as a three-story building. The ground floor holds the marginals, each variable's histogram; the second floor holds the pairwise correlations; and everything from the third floor up holds the higher-order dependence, where any latent types would live, since groups within a dataset show up as conditional dependencies beyond the linear correlation. The copula null rebuilds the ground floor and the second floor exactly as in the real data and leaves every floor above empty: once the correlation matrix is fixed, its Gaussian dependence adds no further joint structure. If clustering finds as many components in a building whose upper floors are empty as it does in the real data, those components were never on the upper floors of the real data either; they were the shape of the lower two all along.

The construction is a single rank transformation:

Algorithm 1. Gaussian-copula matched null.

Input: a real $n \times p$ subsample X .

1. Take the correlation matrix $C = \text{cor}(X)$ and its Cholesky factor L (adding a 10^{-6} ridge to the diagonal if C is not positive definite).
2. Draw Z , an $n \times p$ matrix of independent standard-normal values, and set $Y = ZL$, so that Y carries the same correlations as X .
3. For each variable j , replace column j of Y with the sorted real values of X_j , laid down in the rank order of Y_j .

Output: X^* , whose every marginal matches X exactly, whose covariance matches to within sampling error (the largest correlation it misses is about 0.009), whose dependence is Gaussian, and which has no clustering built in.

In plain terms, the null is a synthetic dataset whose every histogram and every correlation match the real one but which is guaranteed to contain no real groups; if it produces as many types as the real data, the types were the shape of the data, not a feature of the people. I drew sixty such nulls for each domain dataset and fifty for each facet dataset, each at the matched sample size, and ran the full fourteen-model specification curve on every one.

The matched-null tests

Two quantities are read against the null distribution. The first asks whether the data carry any structure beyond their margins and correlations: ΔBIC , the BIC of the best of the fourteen models minus the one-component BIC,⁵ is required to exceed the 95th percentile of the null's ΔBIC (call this H1). The second is the registered test of how many types there are: the real median count across the fourteen models is required to fall inside the null's 95 percent interval for the count to be judged no larger than a typeless population's (H3). H1 is a low bar, certifying only that the data are not Gaussian, and serves as a precondition; H3 is the test that bears on types.

Taxometric analysis

As a second opinion that does not rest on a mixture model's assumptions, I submit each trait's facets (six per NEO-120 trait, four per HEXACO trait) to the two taxometric procedures named in the registration, MAMBAC and MAXEIG,⁶ through RTaxometrics (Ruscio 2023). The Com-

⁵ By the conventional reading a ΔBIC above 10 is very strong evidence for the better-fitting model (Raftery 1995). The test here does not lean on that convention: it compares the real ΔBIC to the null's own distribution rather than to a fixed cutoff.

⁶ MAMBAC (Mean Above Minus Below A Cut) slides a cut along one indicator and plots the mean difference on another; MAXEIG (Maximum Eigenvalue) tracks the largest eigenvalue of the indicators across

parison Curve Fit Index (CCFI) scores the observed curves against curves simulated from explicitly dimensional and explicitly categorical comparison data, returning a number between zero and one: below .45 reads dimensional, above .55 categorical, with an ambiguous band between (Ruscio et al. 2006). I report each trait's CCFI as the average of the two registered procedures and keep the spread between them in view; the L-Mode curve, which the routine also returns, is reported as a supplementary read. The fifty-item set has no facets, so this part covers only the NEO-120 and HEXACO.

Positive-control simulations

A null finding means nothing unless the test can detect types when they are genuinely present, so I ran the whole procedure on synthetic data with known structure, in two regimes (five variables, 4,000 cases, forty nulls per setting). In the dependence regime, two components share identical standard-normal margins on every variable and differ only in the sign of their within-pair correlations, so the type signal lives entirely in the joint structure and nothing leaks into the histograms; the within-component correlation ranges over 0, .30, .50, .70, and .85. In the separation regime, components differ in their means, so the separation does enter the histograms; the centroid separation ranges from 0 to 3 standard deviations. For each setting I ran the specification curve on the synthetic data and on forty copula nulls of it, and counted a detection when the real median count exceeds the null's 97.5th percentile.

ordered subsamples. A genuine category bows these curves into a characteristic peak, while continuous data leave them flat (Ruscio et al. 2006).

The pre-registered decision rule

The rule was set before the hold-out opened; the full text is in the registration. An instrument is credited with real structure only if it clears H1. I then call its structure continuous only if four things hold together: mean within-trait CCFI under .45 with no trait at .55 or above, in both instrument families; fewer than half of respondents confidently placed in any one component, at a posterior above .8; the first split, from one component to two, carrying more than three-quarters of the total BIC gain; and the median-count test H3. The opposite verdict, real types, needs the opposite signs: a CCFI at .55 or above that repeats across instruments, a clear BIC minimum with gaps between component centers, most respondents confidently placed, and a count that holds steady across datasets. The NEO-120 and the fifty-item set are the confirmatory pair and have to agree; HEXACO is robustness only.

For the record, the remaining statistics are these. The count summarized across models is the median of the fourteen; the first-split share is the two-component gain over the total gain, both measured on the best-of-fourteen envelope; and the confident-assignment share is the fraction of respondents whose top posterior exceeds .8 under the selected solution.

Software

R (R Core Team 2024), with `mclust` for the mixtures and `RTaxometrics` for the CCFI; seed 20260620 throughout. Data, code, and the registration are public.

Results

The covariance assumption sets the count

Start with the count itself: it does not settle. Across the fourteen covariance models the number of components BIC keeps runs from one to ten (Figure 1). At the facet level it spans the whole grid in both the NEO-120 and HEXACO; even at the domain level it runs from three to ten in HEXACO and five to ten in the NEO-120. The parameterization means nothing about people, it is a convenience, yet it moves the apparent number of types across almost the entire range anyone might report. The drift has a direction. The richer ellipsoidal models keep fewer components than the plain spherical and diagonal ones, because a flexible covariance can absorb the continuous shape that a rigid one can only chase by adding pieces. So the count is not about the respondents. The model settled it before anyone looked at the data.

The typical number of 'types' matches a typeless null (pre-registered median statistic)

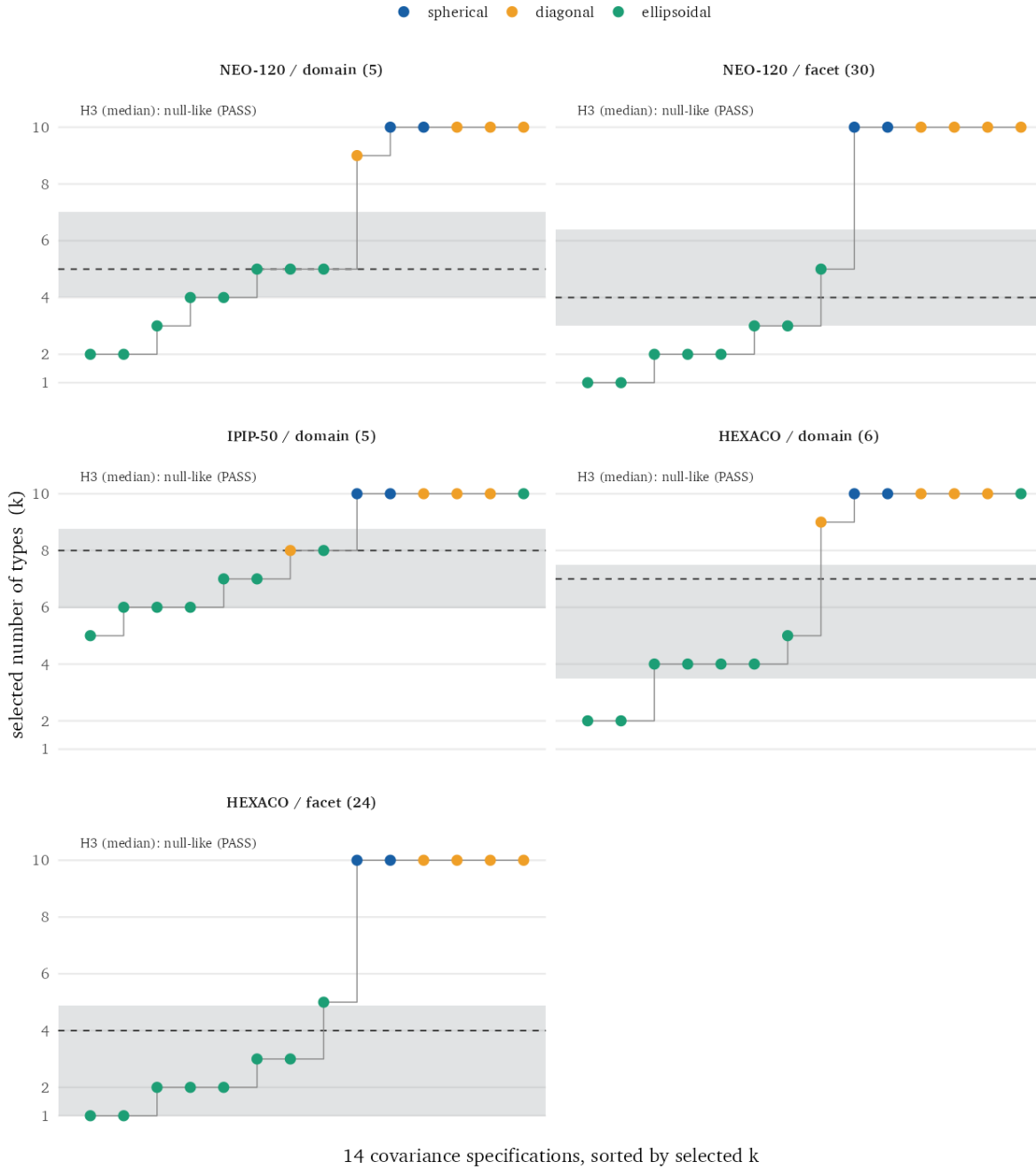


Figure 1: The BIC-selected number of mixture components (“types”) across fourteen covariance parameterizations, on held-out data (matched sample size). Each point is one parameterization, coloured by covariance family and sorted by selected count. The grey band marks the 95% interval of the matched-null median count; the dashed line is the real median. A real line inside the band means the typical count is indistinguishable from data with no latent types.

There is real structure, but weak against the data's own shape

A wandering count does not by itself prove the structure is empty, so I compared each dataset to the matched null. In all five slices the real clustering gain beat the null's 95th percentile: 717 against 485 for the NEO-120 domains, 1,369 against 913 for the fifty-item domains, 816 against 213 for the HEXACO domains, and 4,013 against 1,073 and 2,701 against 337 for the two facet sets. Real personality data, then, carries more than its margins and covariance: some genuine higher-order structure is there in every case. But this mostly certifies that the data are not Gaussian, which was never in doubt; the null fixes only the linear part of the dependence, so beating it is a low bar. The two comparisons that follow, the count and the taxometric curves, are the ones that carry the argument.

The count is no larger than a typeless world's

The next question is whether that real structure shows up as more types. The null has no types by construction, but it still produces multi-component solutions of its own, so the test is whether the real count runs higher. By the measure I registered, the median count across the fourteen models, it does not. The real median sits inside the null's 95 percent interval in every slice: 5 in [4.0, 7.0] for the NEO-120 domains, 4 in [3.0, 6.4] for its facets, 8 in [6.0, 8.8] for the fifty-item domains, 7 in [3.5, 7.5] for the HEXACO domains, and 4 in [1.0, 4.9] for its facets. The grey band in Figure 1 marks the null's interval; wherever the dashed real median falls inside it, the typical number of types is one the typeless world could have produced.

The mean tells a slightly different story, and the difference is worth keeping. Averaged rather than taken at the median, the count does exceed the null in the fifty-item and HEXACO sets. The two

part ways because the per-model counts are skewed: a few covariance models run all the way to the ten-component ceiling, and that tail pulls the mean up while leaving the median where it is. I registered the median, and I report the mean too. That a verdict can hinge on which summary of the same fourteen numbers one takes is, if anything, this paper's own argument turned on itself, and a reason to fix the statistic before looking.

No trait looks like a category

The taxometric test points the same way. No trait reached the categorical threshold in either instrument (Figure 2): across eleven traits the highest was NEO-120 Openness, at .535. The NEO-120 facets averaged .375, the HEXACO facets .430, the latter just below the dimensional cutoff. A few traits sat in the ambiguous band, the highest NEO-120 Openness (.535) and HEXACO Honesty-Humility (.520), but none crossed into categorical territory. One procedure occasionally reached higher, which Figure 2 shows rather than hides; the registered figure, the average of the two, did not.

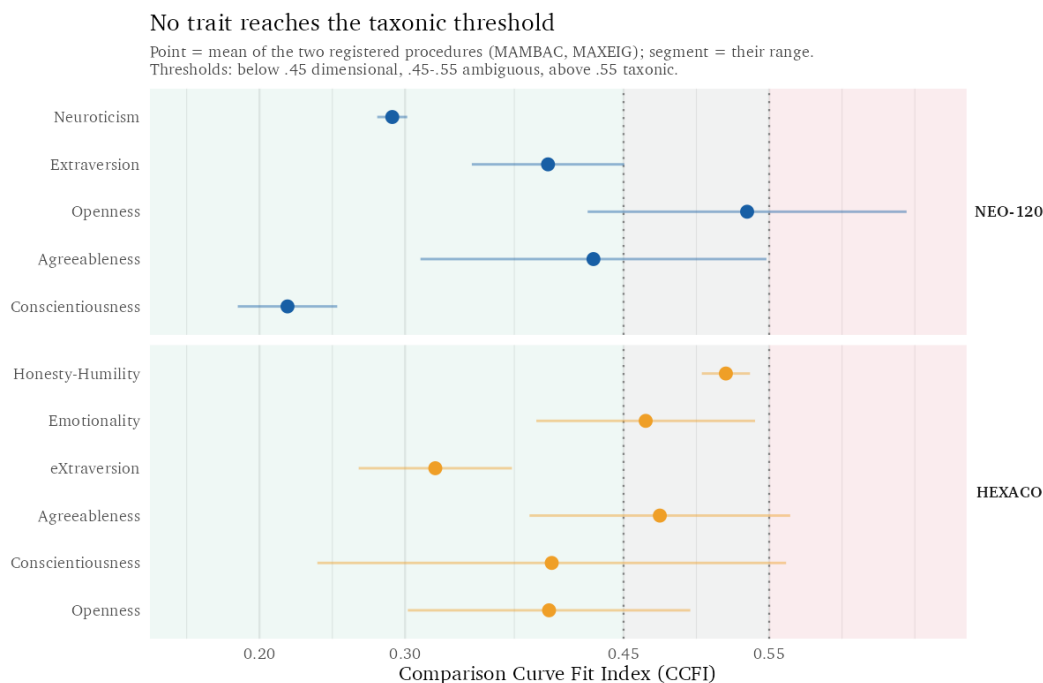


Figure 2: CCFI by trait. Each point is the mean of the two registered procedures, MAMBAC and MAXEIG; the segment is their range. Values below .45 read dimensional, .45 to .55 ambiguous, above .55 taxonic. No trait's mean reaches the taxonic threshold.

People do not sort cleanly, and one cut does most of the work

Two more checks, both registered, lean the same way. Under the selected solution, fewer than half of respondents could be placed in any single component with much confidence, at a posterior above .8: 14.6 and 48.0 percent for the NEO-120 domains and facets, 25.1 percent for the fifty-item set, 30.0 and 46.0 percent for HEXACO. And the first cut, from one component to two, carried more than three-quarters of the total BIC gain in four of the five slices, 88.7 and 89.6 percent for the NEO-120 and 82.8 and 98.9 percent for HEXACO, the lone exception being the fifty-item domains at 61.8 percent. Where the model did split people, it found them spread along the splits, not gathered on one side.

The test can find types when they are there

A failure to find types is worth only as much as the test's power to find them, so I checked that power directly, on synthetic data with a known structure (Figure 3). I built a mixture whose two components share identical marginal distributions and differ only in the orientation of their dependence, so the clustering lives in the joint structure and nowhere in the margins. The test caught it: as the signal grew the real median count climbed out of the null, from one at no signal to nine at a strong one, while the matched null stayed at one throughout. A no-signal version was read, correctly, as typeless. So the test does not only ever return “continuous”: when separated types are present in the dependence, it detects them, more strongly the clearer they are.

It also has a blind spot worth stating plainly. When I instead built well-separated clusters whose separation was large enough to register in the univariate distributions, the test did not flag them: the matched null, which copies those distributions exactly, then produced the same clustering on its own. That is the cost of holding the margins fixed, and here it is the right cost. A type already visible in each variable's histogram is one a continuous distribution with those histograms also produces, so the test declines to call it a discovery. Personality trait distributions are unimodal, so the test operates in the first regime, not the second.

Positive controls: when the test fires, and when it stays quiet

Synthetic data with known types. Grey band = matched-null median-k 95% interval; points = real median-k, red when A: types living only in the dependence are detected, monotonically. B: types separated enough to enter the margins a

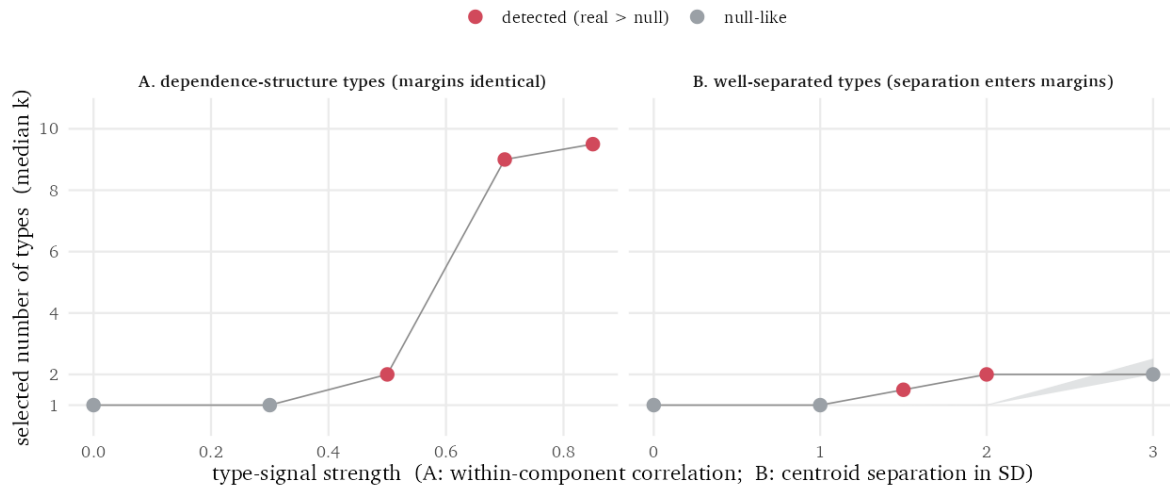


Figure 3: Positive controls on synthetic data with known structure. Grey band: matched-null median count (95%). Points: real median count, red where it exceeds the null. Left, types that live only in the dependence (identical margins, opposite correlation orientation) are detected, and detection grows with the signal. Right, types so separated that the separation enters the margins are not flagged, because the margin-preserving null reproduces them; for unimodal personality data the test operates in the left regime.

The verdict

Held against the rule I set in advance, the categorical signs are missing everywhere: no CCFI at .55 that repeats, no clear BIC minimum, no count that holds still. The NEO-120, the dataset with a clean hold-out, meets all four continuity conditions, and so does HEXACO, whose mean CCFI (.430) sits below the dimensional cutoff. The fifty-item set meets three of the four; it misses only the first-split threshold, at 61.8 percent against 75, and has no facets to give it a CCFI. None of these is a categorical sign. So I find no evidence for separated types in any of the three. The clusters they yield are ones the matched null reproduces, and the mild real structure that remains never turns into extra types or into categories. HEXACO I keep as a robustness check, since its hold-out is not clean.

Discussion

I set out to ask whether the types a mixture model reports in personality data are categories of people or just the data's continuous shape, and I tried to settle it with a test rather than one more clustering. Each dataset met a copula matched null that keeps its margins and covariance; the test ran across all fourteen covariance models; and a taxometric index gave a second reading. Across three large datasets the count ran from one to ten on the covariance assumption alone, sat inside the typeless null by the registered measure in every case, and never reached the taxometric line for a category.

I take this as a result about method, not about people. That trait structure is dimensional is already established (Haslam et al. 2012, 2020); I am less adding to that case than supplying an instrument, one that makes the standing claim "this dataset has N types" something that can be checked, and shows the number at its heart to be set by a choice that means nothing.

The claim is meant to be narrow. The null tests one thing, whether the clustering beats the linear dependence already in the data, and on that test the real data does carry structure the null cannot make; every dataset cleared the bar. But that leftover structure does not become more types, and it does not become taxonic curves. So the honest line is "no evidence for separated types," not "perfectly continuous." Something real is there, in every dataset, below the level of a category, and I do not try to name it here. Skew, mild nonlinearity, some unevenness in density: any of these would do, and none is a kind of person.

One thing in my own analysis makes the point better than I could. Whether the count beats the null depends on whether it is read at the mean or the median, because a few models run to the ceiling and skew the average. I registered the median; the mean says otherwise; I report both. A verdict

that turns on which summary is chosen is exactly the trouble this paper is about, and a reason to name the statistic before the data are in front of you.

The test is not about personality in particular. Any claim that continuous measurements fall into so many groups can be put to a matched null, and the habit of reporting a spread of answers instead of a favorite one travels anywhere a modeling choice is beside the point. It sits alongside taxometrics, which asks the same question from other assumptions, and it builds the constructed null that earlier critics of typology work asked for and did not supply (Katahira et al. 2020); an independence null, again, asks too little.

There are limits. A multiverse is only as honest as the choices it spans, and I varied the covariance assumption fully while holding the mixture family, the selection criterion, and the resolution fixed; a wider one could be drawn. The null tests linear dependence, so a clustering driven by something genuinely curved but still continuous would read as real without being a category, which is close to what I see, and which the null is right not to call a type. The datasets are large but self-selected, three of a kind, all from the IPIP pool, and subsampled to keep the computation in hand; HEX-ACO's hold-out is not clean. And I test for separated categories and their number, not for whether a rough typology is ever a useful way to talk; finding no taxa does not mean a typology can never be a convenient shorthand, only that it is not a discovery of kinds.

The use of all this is small and practical. A number of clusters is not yet a finding. It becomes one only when it beats what the data's own margins and covariance produce, and survives the choices, here the covariance assumption, that should have made no difference. I have left the test where others can pick it up, so that the next claim of N types can be asked, first of all, whether N belongs to people or to the procedure.

Conclusion

The number of types a method reports is not, by itself, a fact about people. For personality, a case studied to the point of consensus, I have shown the count drifting across the whole range as one inconsequential assumption is changed, staying inside what a typeless null produces, and never showing a categorical signature in any trait. None of this is news about personality; the point is the test. It can make a claim of N types fail against a null that means something, and it shows how much the claim leaned on choices nobody was watching. The materials are public, so the next claim can face the same check. Whether a structure is real and whether it is a set of kinds are two questions, and a count of clusters answers neither until it is asked against the right null.

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